Anh Dao

anhdd2000@gmail.com | +358 41 325 1677 | daoa0601.github.io

EDUCATION

M.Sc. (Tech.) in Computer Science, Aalto University — ***Aug 2024 – Jul 2026***
Graduated 31 July 2026. GPA: 4.09/5.00. Minor: Machine Learning, Data Science and AI
Thesis: *Mitigating Sensor-Constrained Attacks on Virtual Reality Cybersickness Classifiers*

B.A. in Computer Science, Grinnell College (GPA 3.76/4.00) — ***Aug 2019 – May 2023***

EXPERIENCE

Smartly – ***Software Engineer*** — ***Feb 2026 – Present***

- Working on the Agentic Orchestration team for Smartly's core initiative Synapse, across **Python**/**Pydantic AI** and **TypeScript**/**Node.js** projects.
- Building Smartly's **Pydantic AI**-based agentic orchestration harness, including progressive tool discovery, Plan Mode control flow and LLM-friendly tool interfaces, as well as Smartly's external MCP server.
- Maintaining **PostgreSQL** storage and **Redis** caching for agents' metadata, message history and draft outputs, plus PoCs for a persisted artifacts log; Redis also backs our **BullMQ** job queues.
- Unblocked 2 core teams through cross-team collaboration, shipping emergency features, bug fixes and refactors.
- Prototyped the company's 1st external MCP gateway and **TypeScript**-based MCP servers, creating the company guidelines on tool design and MCP implementation for 20+ engineering teams.
- Assisted in the 2-person migration of the company's Frontend monorepo of 4 million lines of TypeScript from **Jest** to **Vitest**, and solo-migrated 4 more repos from Jest/ESLint/Prettier to Vitest/Oxlint/Oxfmt.
- Optimized 2 existing LLM-based services through **LangFuse** trace monitoring and evals-based prompt optimization, reducing the cost of inference by 2.5x without loss of performance.
- Part of the company's AI Productivity Frontier group, delivering Knowledge Sharing sessions to 200+ R&D employees.

Musixmatch – ***Research Assistant*** — ***Feb 2026 – Aug 2026***

- Built an ETL data pipeline in Python to exhaustively collect, clean, and deduplicate Musixmatch's lyrics dataset through their provisioned API, up to 10M+ tracks.
- Ran batch LLM annotation on 15,000+ Finnish tracks, cleaning up malformed LLM outputs into structured data.
- Replicated and extended prior work on Finnish hip-hop networks and rhyme analysis with cross-linguistic and literary analysis. Link.

Aalto University – ***Master's Thesis Worker*** — ***Jan 2026 – Jul 2026***

- Developed feature-constrained gradient attacks and a confidence-triggered fallback defense for VR cybersickness classifiers using sensor-based LSTM models.

Nokia Solutions & Networks – ***R&D Trainee*** — ***Aug 2025 – Jan 2026***

- Built a **Python** ETL pipeline indexing 100,000+ documents into **Vertex AI Search** on **GCP**, unifying product search for a RAG engine.
- Prototyped a multi-agent harness with Gemini Agent Platform combining fanout research, network logs triaging, and RAG-based QnA, reducing incident triage time by ∼40%; built the supporting LLM tools in **Python** and **React**.

Aalto University – ***Teaching Assistant*** — ***Aug 2024 – 2025***

- Assisted in core Master's-level Security courses, totalling over 500 students, by holding office hours, designing network security assignments, and creating tools for fast grading.

Grinnell College (DASIL) – ***Research Assistant*** — ***Aug 2022 – May 2023***

- Built multivariate statistical models in **R** and **Python** from Stat2Labs game data (platform used by 1,000+ students across 5 institutions).

Microsec Zrt. – ***Software Engineer Trainee*** — ***Jun – Aug 2022***

- Extended a PKI certificate linter in **Go** to handle X.509 extension edge cases following RFC 5280.

PROJECTS

Teleport Contest – Autonomous Coding Harness ***2026 – Present***

- Participating in the Teleport Coding Challenge, using an agent swarm to port NetHack 5.0, a complex roguelike game with 500,000 lines of C and Lua code, to JavaScript with byte-exact behavior.
- Designing guardrails, multi-agent orchestration, evaluation mechanisms to prevent common pitfalls with independent long-horizon agents like reward hacking, challenge bypass and goal misalignment.

MLX Silicon Inference assorted projects ***2026***

- Ported Open Source Decision models into an MLX-compatible framework, including Cloudflare Clef. Evaluated and optimized inference performance on an M4 Pro with relevant Jev-based evals and benchmarks.
- In separate **llama.cpp** experiments using **Metal**, speculative decoding increased Gemma 4 12B throughput by 12.1% across three tasks.

Scalable Agentic Systems Evaluation ***Jan – Apr 2026***

- Ran SWE-Bench Lite evaluations on 300 software engineering tasks, utilizing open source models like GLM-4.7, across different harnesses like mini-swe-agent and Claude Code
- Deployed **vLLM** with **ROCm** on LUMI's AMD MI250X GPUs using **SLURM**; profiled GPU utilization, memory, latency, and concurrency for 6+ open-weight LLMs on these same tasks, fixing incompatible inference formats and missing support for AMD-based inference cookbooks.